ASSIGNMENT-2

Category	Details	
Aim	To practice and implement control structures, specifically **if-else statements** and **for loops**, in Python[2].	
Tasks to be Done		**Task 1:** Check if an input number is even or odd[3]. **Task 2:** Calculate the sum of integers from 1 to 50 using a loop[4].
Requirements	Python environment, GitHub repository for submission[55].	
Objectives	Understand and correctly use the modulo operator (%), implement conditional logic with if-else, and utilize for loops for iteration and accumulation[666].	
Submission	A link to a GitHub repository containing two Python scripts (one for each task) and a README.md file describing their functionality[7].	

Task 1: Check if a Number is Even or Odd

Theory

An **even number** is an integer that is exactly divisible by 2, leaving a remainder of 0. An **odd number** is an integer that is not exactly divisible by 2, leaving a remainder of 1.

In Python, the **modulo operator** (%) is used to find the remainder of a division. The condition to check for an even number is number % 2 == 0[8]. The **if-else statement** is a control structure used to execute different blocks of code based on whether a condition is True or False[9].

Algorithm

1. **Start.**

2. Prompt the user to **input an integer** and store it in a variable (e.g., num)[10].
3. Use an **if-else statement** to check the condition:
    - **IF** the remainder of num divided by 2 is 0 (num % 2 == 0)[11]:
        - Display that the number is **even**[1212].
    - **ELSE** (the remainder is not 0):
        - Display that the number is **odd**[1313].
4. **End.**

Code

Save the following code as **task1_even_odd.py**:

Python

```python
# Task 1: Check if a Number is Even or Odd
# Takes an integer input and uses an if-else statement to determine if it's even or odd.

try:
    # 1. Takes an integer input from the user.
    number = int(input("Enter a number: "))

    # 2. Checks whether the number is even or odd using an if-else statement.
    if number % 2 == 0:
        # 3. Displays the result accordingly.
        print(f"{number} is an even number.")
    else:
        # 3. Displays the result accordingly.
        print(f"{number} is an odd number.")

except ValueError:
```

```python
    print("Invalid input. Please enter a whole number (integer).")
```

Task 2: Sum of Integers from 1 to 50 Using a Loop

Theory

Iteration is the process of repeating a set of instructions. A **for loop** is an iteration control structure in Python that is used to iterate over a sequence (like a range of numbers)[14].

The built-in Python function **range(start, stop)** is used to generate a sequence of numbers. To include numbers from 1 up to and including 50, the loop should use range(1, 51), as the stop value is exclusive.

Accumulation involves initializing a variable (e.g., total_sum = 0) and repeatedly adding a value to it during each iteration of the loop[15].

Algorithm

1. **Start.**
2. Initialize a variable, total_sum, to **0** to store the running sum[16].
3. Use a **for loop** to iterate through the numbers from **1 to 50** (using range(1, 51))[17].
4. In each iteration, **add the current number** to the total_sum[18].
5. After the loop finishes, **display the final total_sum**[1919].
6. **End.**

Code

Save the following code as **task2_sum_of_numbers.py**:

Python

```python
# Task 2: Sum of Integers from 1 to 50 Using a Loop

# Uses a for loop to calculate the sum of all integers from 1 to 50.


# Initialize the accumulator variable

total_sum = 0
```

```
# 1. Uses a for loop to iterate over numbers from 1 to 50.
# range(1, 51) includes 1 and goes up to (but not including) 51.
for number in range(1, 51):
   # 2. Calculates the sum of all integers in this range.
   total_sum += number  # Equivalent to total_sum = total_sum + number


# 3. Displays the final sum.
# Expected Output: The sum of numbers from 1 to 50 is: 1275
print(f"The sum of numbers from 1 to 50 is: {total_sum}")
```

Learning Outcomes

By completing this assignment, you will have demonstrated proficiency in the following:

- **Integer Input:** Handling user input and converting it to the correct data type (int)[20].
- **Conditional Logic (if-else):** Implementing decision-making logic using the if-else control structure[21].
- **Arithmetic Operators:** Correctly using the **modulo operator (%)** for divisibility checks.
- **Iteration (for loop):** Using a for loop and the range() function to iterate over a defined sequence of numbers[22].
- **Accumulation:** Employing a variable as an **accumulator** to aggregate a result (the sum) across iterations[23].
- **Program Output:** Formatting and displaying clear output to the user[24242424].
- **Version Control:** Utilizing **GitHub** for code storage and submission[25252525].

CODE-2

```
# Task 1: Check if a Number is Even or Odd
# Takes an integer input and uses an if-else statement to determine if it's even or odd.
```

```
try:
    # 1. Takes an integer input from the user[cite: 5].
    # We use int(input(...)) to ensure the input is an integer.
    number = int(input("Enter a number: "))

    # 2. Checks whether the number is even or odd using an if-else statement[cite: 6].
    # A number is even if the remainder when divided by 2 is 0.
    if number % 2 == 0:
        # 3. Displays the result accordingly[cite: 7].
        print(f"{number} is an even number.")
    else:
        # 3. Displays the result accordingly[cite: 7].
        print(f"{number} is an odd number.")

except ValueError:
    print("Invalid input. Please enter a whole number (integer).")
```

User Input	Program Output
7	Enter a number: 7 7 is an odd number.
12	Enter a number: 12 12 is an even number.

CODE-3

```
# Task 2: Sum of Integers from 1 to 50 Using a Loop
# Uses a for loop to calculate the sum of all integers from 1 to 50.

# Initialize the accumulator variable to store the running sum.
```

```python
total_sum = 0

# 1. Uses a for loop to iterate over numbers from 1 to 50[cite: 12].
# range(1, 51) starts at 1 and stops before 51, including 50.
for number in range(1, 51):
    # 2. Calculates the sum of all integers in this range[cite: 13].
    total_sum += number  # Add the current number to the running total

# 3. Displays the final sum[cite: 14].
print(f"The sum of numbers from 1 to 50 is: {total_sum}")
```

```
The sum of numbers from 1 to 50 is: 1275 [cite: 15]
```